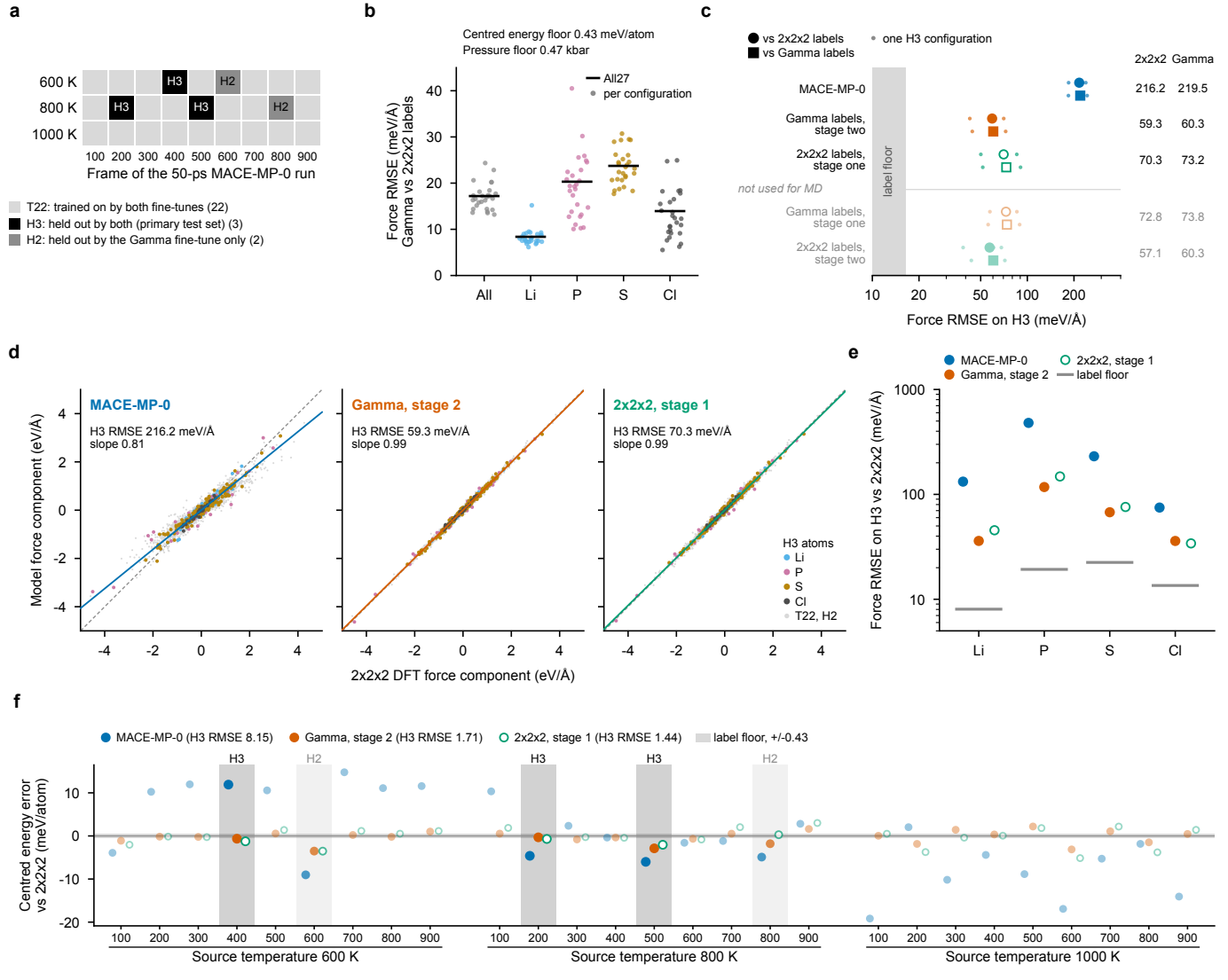

a
600 K
800 K
1000 K
H3
H2
100 200 300 400 500 600 700 800 900
Frame of the 50-ps MACE-MP-0 run
T22: trained on by both fine-tunes (22)
H3: held out by both (primary test set) (3)
H2: held out by the Gamma fine-tune only (2)
b
Centred energy floor 0.43 meV/atom
Pressure floor 0.47 kbar
All27
per configuration
Force RMSE (meV/Å)
Gamma vs 2x2x2 labels
All Li P S Cl
c
vs 2x2x2 labels
vs Gamma labels
one H3 configuration
2x2x2 Gamma
MACE-MP-0 216.2 219.5
Gamma labels, stage two 59.3 60.3
2x2x2 labels, stage one 70.3 73.2
not used for MD
Gamma labels, stage one 72.8 73.8
2x2x2 labels, stage two 57.1 60.3
label floor
Force RMSE on H3 (meV/Å)
d
MACE-MP-0
H3 RMSE 216.2 meV/Å
slope 0.81
Gamma, stage 2
H3 RMSE 59.3 meV/Å
slope 0.99
2x2x2, stage 1
H3 RMSE 70.3 meV/Å
slope 0.99
H3 atoms
Li
P
S
Cl
T22, H2
Model force component (eV/Å)
2x2x2 DFT force component (eV/Å)
e
MACE-MP-0
2x2x2, stage 1
Gamma, stage 2
label floor
Force RMSE on H3 vs 2x2x2 (meV/Å)
Li P S Cl
f
MACE-MP-0 (H3 RMSE 8.15)
Gamma, stage 2 (H3 RMSE 1.71)
2x2x2, stage 1 (H3 RMSE 1.44)
label floor, +/-0.43
H3
H2
Centred energy error
vs 2x2x2 (meV/atom)
Source temperature 600 K
Source temperature 800 K
Source temperature 1000 K